

SQL Homework 2

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3.1 查询 1991 年出生的读者姓名、工作单位和身份证号。

```
SELECT readerName, workUnit, identitycard
FROM dbo.Reader
WHERE SUBSTRING(identitycard, 7, 4) = '1991';
```

	readerName	workUnit	identitycard
1	陈辉	南昌市电脑研制公司	010207199111014221

3.2 查询图书名中含有“数据库”的图书的详细信息。

```
SELECT *
FROM dbo.Book
WHERE bookName LIKE '%数据库%';
```

	bookNo	classNo	bookName	authorName	publisherNo	price	publishingDate	shopDate	shopNum
1	B201503001	C003	数据库系统原理	万慧红	P002	22.20	2014-01-01 00:00:00.000	2015-01-11 00:00:00.000	8
2	B201503002	C003	数据库系统概念	王珊	P005	28.20	2014-05-01 00:00:00.000	2015-01-11 00:00:00.000	5

3.3 查询在 2015—2016 年之间入库的图书编号、出版时间、入库时间和图书名称，并按入库时间的降序排序输出。

```
SELECT bookNo, publishingDate, shopDate, bookName
FROM dbo.Book
WHERE shopDate BETWEEN '2015-01-01' AND '2016-12-31';
ORDER BY shopDate DESC;
```

	bookNo	publishingDate	shopDate	bookName
1	B201603003	2016-10-02 00:00:00.000	2016-11-12 00:00:00.000	数据结构与算法
2	B201603001	2016-01-02 00:00:00.000	2016-07-12 00:00:00.000	操作系统教程
3	B201605001	2015-09-02 00:00:00.000	2016-04-12 00:00:00.000	现代诗歌欣赏
4	B201601004	2015-01-02 00:00:00.000	2016-01-12 00:00:00.000	微观经济学
5	B201501002	2014-07-01 00:00:00.000	2016-01-11 00:00:00.000	大学英语
6	B201506001	2015-06-01 00:00:00.000	2016-01-11 00:00:00.000	大学英语听说
7	B201601003	2015-03-02 00:00:00.000	2015-12-12 00:00:00.000	数据结构
8	B201605002	2015-02-02 00:00:00.000	2015-08-12 00:00:00.000	现代操作系统
9	B201603004	2014-11-02 00:00:00.000	2015-01-12 00:00:00.000	古典文学作品欣赏
10	B201601001	2014-01-02 00:00:00.000	2015-01-12 00:00:00.000	操作系统原理
11	B201601002	2014-01-02 00:00:00.000	2015-01-12 00:00:00.000	大学英语读写实践
12	B201501001	2014-10-01 00:00:00.000	2015-01-11 00:00:00.000	政治经济学
13	B201501003	2014-10-01 00:00:00.000	2015-01-11 00:00:00.000	宏观经济学
14	B201503001	2014-01-01 00:00:00.000	2015-01-11 00:00:00.000	数据库系统原理
15	B201503002	2014-05-01 00:00:00.000	2015-01-11 00:00:00.000	数据库系统概念

3.4 查询读者“喻自强”借阅的图书编号、图书名称、借书日期和归还日期。

```
SELECT
    borrow.bookNo,
    book.bookName,
    borrow.borrowDate,
    borrow.returnDate
FROM
    dbo.Borrow AS borrow
JOIN dbo.Reader AS reader ON borrow.readerNo = reader.readerNo
JOIN dbo.Book AS book ON borrow.bookNo = book.bookNo
WHERE reader.readerName = '喻自强';
```

	bookNo	bookName	borrowDate	returnDate
1	B201501001	政治经济学	2016-03-11 00:00:00.000	2016-10-11 00:00:00.000
2	B201601001	操作系统原理	2016-09-17 00:00:00.000	2016-11-05 00:00:00.000
3	B201601002	大学英语读写实践	2016-03-01 00:00:00.000	2016-10-11 00:00:00.000
4	B201601004	微观经济学	2016-03-11 00:00:00.000	2016-08-11 00:00:00.000
5	B201603004	古典文学作品欣赏	2016-05-01 00:00:00.000	2016-10-11 00:00:00.000

3.5 查询借阅了清华大学出版社出版的图书的读者编号、读者姓名、图书名称、借书日期和归还日期。

```
SELECT
    r.readerNo,
    r.readerName,
    b.bookName,
    bw.borrowDate,
    bw.returnDate
FROM
    dbo.Reader AS r
JOIN dbo.Borrow AS bw ON r.readerNo = bw.readerNo
JOIN dbo.Book AS b ON b.bookNo = bw.bookNo
JOIN dbo.Publisher AS p ON b.publisherNo = p.publisherNo
WHERE p.publisherName = '清华大学出版社';
```

	readerNo	readerName	bookName	borrowDate	returnDate
1	R2016003	张晓梅	数据库系统原理	2016-01-07 00:00:00.000	NULL
2	R2016004	张良	数据结构	2016-12-07 00:00:00.000	NULL

3.6 查询会计学院没有归还所借图书的读者编号、读者姓名、图书名称、借书日期和应归还日期。

```
SELECT
    r.readerNo,
    r.readerName,
    bk.bookName,
    bw.borrowDate,
    bw.shouldDate
FROM
    dbo.Book as bk
JOIN dbo.Borrow AS bw ON bk.bookNo = bw.bookNo
JOIN dbo.Reader AS r ON r.readerNo = bw.readerNo
JOIN dbo.BookClass AS bc on bc.classNo = bk.classNo
WHERE bc.classNo = 'C001'
    AND bw.returnDate IS NULL;
```

	readerNo	readerName	bookName	borrowDate	shouldDate
1	R2015001	张小娟	政治经济学	2015-10-01 00:00:00.000	2015-11-01 00:00:00.000
2	R2015002	刘凤	微观经济学	2016-11-07 00:00:00.000	2016-12-07 00:00:00.000
3	R2016001	张露	政治经济学	2016-09-01 00:00:00.000	2016-12-01 00:00:00.000
4	R2016001	张露	微观经济学	2016-09-07 00:00:00.000	2016-10-07 00:00:00.000

3.7 查询在 2015—2016 年之间借阅但还未归还图书的读者编号、读者姓名以及这些借阅未归还图书的图书编号、图书名称和借书日期。

```
SELECT
    r.readerNo,
    r.readerName,
    bk.bookNo,
    bk.bookName,
    bw.borrowDate
FROM
    dbo.Book as bk
JOIN dbo.Borrow AS bw ON bk.bookNo = bw.bookNo
JOIN dbo.Reader AS r ON r.readerNo = bw.readerNo
WHERE bw.returnDate IS NULL
    AND YEAR(bw.borrowDate) BETWEEN '2015' AND '2016';
```

	readerNo	readerName	bookNo	bookName	borrowDate
1	R2015001	张小娟	B201501001	政治经济学	2015-10-01 00:00:00.000
2	R2015002	刘凤	B201601004	微观经济学	2016-11-07 00:00:00.000
3	R2016001	张露	B201501001	政治经济学	2016-09-01 00:00:00.000
4	R2016001	张露	B201601001	操作系统原理	2016-05-07 00:00:00.000
5	R2016001	张露	B201601004	微观经济学	2016-09-07 00:00:00.000
6	R2016003	张晓梅	B201503001	数据库系统原理	2016-01-07 00:00:00.000
7	R2016003	张晓梅	B201601001	操作系统原理	2016-09-07 00:00:00.000
8	R2016004	张良	B201601001	操作系统原理	2016-12-07 00:00:00.000
9	R2016004	张良	B201601003	数据结构	2016-12-07 00:00:00.000
10	R2016004	张良	B201603001	操作系统教程	2016-04-17 00:00:00.000

3.8 查询每种类别图书的分类号、分类名称、最高价格和平均价格，并按最高价格的升序输出。

```
SELECT
    bc.classNo,
    bc.className,
    MAX(bk.price) AS MaxPrice,
    AVG(bk.price) AS AvgPrice
FROM dbo.BookClass AS bc
JOIN dbo.Book AS bk ON bc.classNo = bk.classNo
GROUP BY bc.classNo, bc.className
ORDER BY MaxPrice ;
```

	classNo	className	MaxPrice	AvgPrice
1	C005	文学类	21.00	19.600000
2	C002	外语类	25.20	20.866666
3	C001	经济类	30.30	27.200000
4	C003	计算机类	41.00	25.614285

3.9 查询每个读者在借（即借阅未归还）的图书数量、读者编号、读者姓名和工作单位，并按借书数量的降序排序输出。

```
SELECT
    COUNT(*) AS BorrowNum,
    r.readerNo,
    r.readerName,
    r.workUnit
FROM dbo.Reader AS r
JOIN dbo.Borrow AS bw ON r.readerNo = bw.readerNo
WHERE bw.returnDate IS NULL
GROUP BY r.readerNo, r.readerName, r.workUnit
ORDER BY BorrowNum DESC;
```

	BorrowNum	readerNo	readerName	workUnit
1	3	R2016001	张露	上海生物研究室
2	3	R2016004	张良	上海生物研究室
3	2	R2016003	张晓梅	统一股份有限公司
4	1	R2015001	张小娟	统一股份有限公司
5	1	R2015002	刘凤	联合股份有限公司

3.10 查询每个出版社出版的每种类别的图书平均价格，要求显示出版社名称、图书类别名称和平均价格。

```
SELECT
    p.publisherName,
    bc.className,
    AVG(bk.price) AS AvgPrice
FROM dbo.Publisher AS p
JOIN dbo.Book AS bk ON p.publisherNo = bk.publisherNo
JOIN dbo.BookClass AS bc ON bk.classNo = bc.classNo
GROUP BY p.publisherName, bc.className;
```

	publisherName	className	AvgPrice
1	高等教育出版社	计算机类	23.550000
2	机械工业出版社	计算机类	25.100000
3	清华大学出版社	计算机类	27.333333
4	中国人民大学出版社	经济类	27.200000
5	外语教学与研究出版社	外语类	18.700000
6	中国人民大学出版社	外语类	25.200000
7	中国人民大学出版社	文学类	19.600000

3.11 查询在借图书的总价不低于 200 元的读者编号、读者姓名和在借图书总价。

```
SELECT
    r.readerNo,
    r.readerName,
    SUM(bk.price) AS TotalPrice
FROM dbo.Reader AS r
JOIN dbo.Borrow AS bw ON r.readerNo = bw.readerNo
JOIN dbo.Book AS bk ON bw.bookNo = bk.bookNo
WHERE bw.returnDate IS NULL
GROUP BY r.readerNo, r.readerName
HAVING SUM(bk.price) >= 200;
```

不存在这样的读者